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ASEAS: A Transformer-Based OCR and Retrieval-Augmented Framework for Automated Handwritten Academic Script Evaluation

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Automated evaluation of handwritten academic scripts remains a challenging problem due to variability in handwriting styles and the complexity of subjective grading. This paper presents ASEAS (Automated Script Evaluation and Analysis System), a multi-stage AI-based framework that integrates Transformer-based Optical Character Recognition (OCR), semantic embedding models, Retrieval-Augmented Generation (RAG), and rubric-constrained Large Language Models for automated academic assessment.

The proposed system first performs high-accuracy handwritten text recognition using a fine-tuned Transformer OCR module. The extracted text is then processed through a retrieval-enhanced semantic evaluation pipeline that aligns responses with rubric expectations and reference content. Experimental analysis on pilot-scale assignment datasets demonstrates transcription accuracy exceeding 94% and strong agreement with human evaluators (overall Pearson correlation ≈ 0.87). Ablation studies further confirm the importance of contextual retrieval and rubric grounding in improving grading consistency.

The results indicate that ASEAS provides a scalable, consistent, and time-efficient alternative to manual grading, offering approximately threefold reduction in evaluation time while maintaining strong semantic alignment with human assessment standards.

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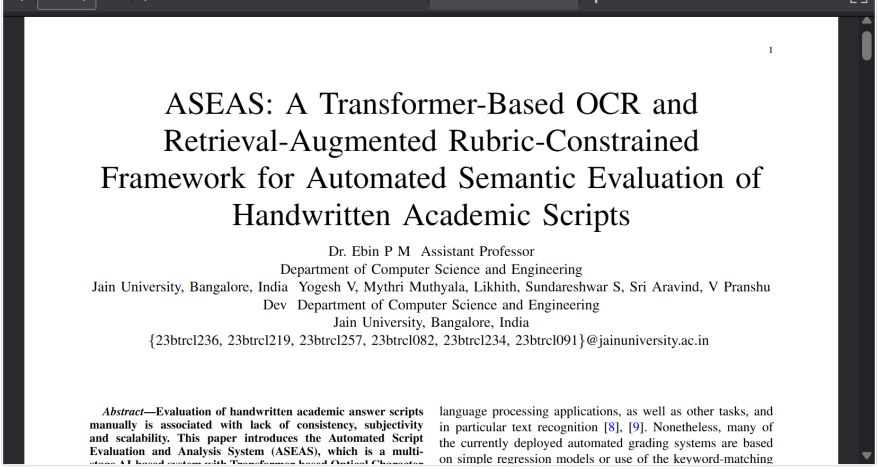

ASEAS: A Transformer-Based OCR and Retrieval-Augmented Rubric-Constrained Framework for Automated Semantic Evaluation of Handwritten Academic Scripts

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Abstract—Evaluation of handwritten academic answer scripts manually is associated with lack of consistency, subjectivity and scalability. This paper introduces the Automated Script Evaluation and Analysis System (ASEAS), which is a multi-

language processing applications, as well as other tasks, and in particular text recognition [8], [9]. Nonetheless, many of the currently deployed automated grading systems are based on simple regression models or use of the keyword-matching

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